Md. Rezuwan Hassan

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Research Interests

Robotics and systems engineering; Low-resource NLP and speech technology; regional dialects; Broader interests include language-model applications for underrepresented languages.

Education

M.S. in Computer Science and Engineering, BRAC University — October 2021 – 2024
Major: Artificial Intelligence — CGPA: 3.88/4.00
Thesis: *A character gram modeling approach towards Bengali Speech to Text with Regional Dialects*.

B.S. in Electrical and Electronic Engineering, BRAC University — January 2016 – April 2020
Major: Communication and Wireless Technologies — CGPA: 3.45/4.00
Thesis: *Efficient approach for reliability evaluation of distribution system considering momentary interruption*.

Research Experience

Graduate Research Assistant (AI, NLP & HCI), BRAC University — February 2024 – October 2025

- Designed research protocols, experimental pipelines, and methodologies; documented findings for research publications.
- Processed, analyzed, curated, and developed datasets from multiple sources and data modalities, including Bengali speech research.
- Automated research workflows and fine-tuned deep-learning models to support project-specific experimental objectives.
- Co-supervised and evaluated undergraduate thesis students, providing technical guidance during their research projects.

Publications and Research Outputs

Md. Rezuwan Hassan is bolded. Author order follows the cited records.

Peer-reviewed conference and workshop papers

1. **Are ASR foundation models generalized enough to capture features of regional dialects for low-resource languages?**
   Tawsif Tashwar Dipto, Azmol Hossain, Rubayet Sabbir Faruque, **Md. Rezuwan Hassan**, Kanij Fatema, Tanmoy Shome, Ruwad Naswan, Md.Foriduzzaman Zihad, Mohaymen Ul Anam, Nazia Tasnim, Hasan Mahmud, Md. Kamrul Hasan, Md. Mehedi Hasan Shawon, Farig Sadeque, and Tahsin Reasat. In *Proceedings of the 14th International Joint Conference on Natural Language Processing and the 4th Conference of the Asia-Pacific Chapter of the Association for Computational Linguistics*, 2025, pp. 178–188.

2. **Meaning Over Morphology: A Multi-Metric Benchmark of LLMs for Bangla Dialect Translation**
   Soumik Deb Niloy, Subhey Sadi Rahman, Mahbub E Sobhani, Md. Golam Rabiul Alam, Farig Yousuf Sadeque, and **Md. Rezuwan Hassan**. In *Proceedings of the First Workshop on Dialects in NLP – A Resource Perspective* (DialRes), 2026, pp. 238–255.

3. **A Horizontal Federated Random Forest for Heart Disease Detection from Decentralized Local Data**
   Shafin Mahmud, **Md. Rezuwan Hassan**, Md. Ashfaqul Haque, and Md. Golam Rabiul Alam. In *2022 IEEE 10th Region 10 Humanitarian Technology Conference* (R10-HTC).

4. **Monte Carlo Simulation for Reliability Worth Assessment of Distribution System Considering Momentary Interruptions**
   Md. Tanjil Ahmed, **Md. Rezuwan Hassan**, Palash Chandra Ghosh, Mohammad Saiful Huq, Mohaimenul Islam, and A. S. Nazmul Huda. In *2022 International Conference on Energy and Power Engineering* (ICEPE).

Preprints

1. **RegSpeech12: A Regional Corpus of Bengali Spontaneous Speech Across Dialects**
   Md. Rezuwan Hassan, Azmol Hossain, Kanij Fatema, Rubayet Sabbir Faruque, Tanmoy Shome, Ruwad Naswan, Trina Chakraborty, Md. Foriduzzaman Zihad, Tawsif Tashwar Dipto, Nazia Tasnim, Nazmuddoha Ansary, Md. Mehedi Hasan Shawon, Ahmed Imtiaz Humayun, Md. Golam Rabiul Alam, Farig Sadeque, and Asif Sushmit. arXiv, 2025.

Manuscripts under review

1. **RoshikBot: Towards Culturally Grounded Bangla Satire and Humor with Foundation Models**
   Nusrat Jahan Mim, Md. Ishmam Tasin, **Md. Rezuwan Hassan**, Tanmoy Shome, Zumaina Islam, Syed Ishtiaque Ahmed, Farida Chowdhury, and S M Taiabul Haque. Under review at the ACM CHI Conference on Human Factors in Computing Systems (CHI), 2027.

Technical report

1. **AgriBuddy: An AI-powered Agent-based System for Bangladeshi Agriculture Using RAG and Vision Models**
   Md. Shaleh Islam Tonmoy, Rawhatur Rabbi, **Md. Rezuwan Hassan**, Ruwad Naswan, and Tanmoy Shome. Technical report, 2025.

   Overview Video | Presentation Slides | Technical Report

Selected Research Projects

Bengali Speech Recognition with Regional Dialects Master's thesis project
Built a regional speech corpus and fine-tuned Wav2Vec2 for Bengali dialect transcription; subsequently fine-tuned a Whisper model and deployed it as a Hugging Face demo.
Code | Thesis report | Demo | Model weights

Bengali dialect and speech-to-IPA resources
Contributed to Bengali.AI speech and dialect projects, including regional speech-to-IPA, dialect-to-IPA transcription, and dialect standardization.

Automatic Bangla Longform Audio Transcription with Diarization
Whisper-based long-form transcription project that produces speaker-diarized Bangla transcripts, English translation, and CSV/Excel outputs.

Technical Skills

Programming: Python, Dart, Embedded C, SQL
Machine learning: TensorFlow, scikit-learn, Hugging Face, Fast.ai, ONNX
NLP and AI: NLP, ASR, model fine-tuning, RAG, NLTK, Librosa, OpenAI, Gemini, Claude
EEE & Engineering Tools: MATLAB/Simulink, PSpice, LTspice, Proteus, Arduino, Verilog, SysModeler
Data and engineering: NumPy, Pandas, OpenCV, GitHub, VS Code, Google Colab, Kaggle, Streamlit

Awards and Honors

Merit-based master's scholarship, BRAC University 2022–2023
Full tuition waiver awarded based on academic performance.

Vice Chancellor's List, BRAC University Spring 2020
Recognized after earning a 4.00 GPA across 18 credits.

Team runner-up, National AI Hackathon, Bangladesh Innovation Conclave 2025
AgriBRACUion team result; AgriBuddy project. Presentation

Teaching, Mentoring, and Academic Service

Instructor and course developer, Lead Academy October–December 2023
Developed and delivered Natural Language Processing (NLP) for Beginners using Python.

Workshop instructor and mentor: NLP Datathon Bootcamp; National AI Workshop on Speech Recognition; invited workshops on Bengali language and AI.

Datathon judge: invited judge at events hosted by BUET, University of Dhaka, IUT, and UIU.

Industry Experience

Senior Technical Project Manager, SysModeler Inc. July 2026–present
Coordinate technical delivery, review work and defect resolution, translate requirements into tasks, and maintain project documentation.

AI Engineer, SysModeler Inc. March 2025–June 2025
Developed NLP system-comprehension capabilities using RAG and vision models; built a computer-vision pipeline to convert system-skeleton images into SysML v1 diagrams; investigated requirements-extraction limitations.

References

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